Tutorial 5

The core exercises are considered part of the course material, and you are advised to at least finish these. For additional practice and interesting applications, advanced exercises are available as well – feel free to pick and choose from these.

Unless otherwise specified, we expect you to create your own algorithms for the exercises in each tutorial. This means that you should not use libraries (e.g. SciPy) for the methods that are part of the course material.

Core exercises

1. **Differentiation**
   Let's consider the following function:

$$f(x) = x^2 \sin(x). \tag{1}$$

   (a) Calculate the analytical derivative of this function and plot the function in the range $[0, 2\pi]$.

   (b) Make a function to evaluate the derivative in the same interval using the central difference method, calculated at 200 points evenly spaced. Do this for $h = 1, 0.1, 0.01$ and 0.001. Compare the different h values and compare with the analytical derivative.

   (c) Make a differentiator using Ridder's method to evaluate the same 200 x-values as above, use 3 different target accuracies. Which initial h values yield stable results? Why? Compare the results with part (b).

2. **Generating Random Numbers on a Sphere**

   (a) In a spherical coordinate system, any point can be expressed in the units of (r, θ, ϕ). If we consider a unit sphere (i.e. $r = 1$), a random point is equivalent to a random direction characterized by random generation of θ and ϕ. Firstly, generate random points on a unit sphere using the following description of the angles:

$$\theta = \pi P_u(0, 1) \tag{2}$$
$$\phi = 2\pi P_u(0, 1) \tag{3}$$

   Here P_u is a uniform random number generator. Use `numpy.random` to generate random values and plot 5000 random points in a 3D plot.

   (b) Now use a different definition for randomizing the angles as follows and repeat the above procedure:

$$\theta = \cos^{-1}(1 - 2P_u(0, 1)) \tag{4}$$
$$\phi = 2\pi P_u(0, 1) \tag{5}$$

   (c) Compare the two plots. Which one is better? Why? Explain how to correctly place random numbers on a sphere and why this holds. Hint: Use the transformation method and remember that the area of each surface element on a unit sphere is $\sin(\theta)\mathrm{d}\theta\mathrm{d}(\phi)$.

Advanced exercises

3. **Linear Congruential generator**
   In astrophysics as a whole it is of essential importance to generate good pseudo-random numbers in order to make initial conditions or do Monte Carlo simulations. In this part of the tutorial we will

construct a linear congruential generator. In general, a linear congruential generator is given by the following recursive relation:

$$X_{n+1} = (aX_n + c) \bmod m. \tag{6}$$

The first obvious step is comparing the relation of the outcome of the nth number with the $(n+1)$th outcome. In this exercise we will exactly do that for different situations. A mathematical measure to calculate the correlation between two arrays of numbers is called the Pearson correlation coefficient. Calculate the Pearson correlation coefficient for your random numbers:

$$r_{\mathrm{xy}} = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\sqrt{\mathrm{var}(x)}\sqrt{\mathrm{var}(y)}}, \tag{7}$$

where x and y are the arrays of numbers for which we want to calculate the correlation. In our case use x_i and x_{i+1}. Choose a seed which is identical to that of one of your classmates in this exercise.

(a) Choose a small number a and a small prime for m. Now plot the comparison with n and $n+1$, also calculate the Pearson correlation coefficient. What will be a possible disadvantage of using a prime modulus? And what will be an advantage?

(b) Choose a small number a and a power of two like 2^{32} and make the same plot as before and calculate the correlation. What can you say about this plot? What will be a possible disadvantage of using a power of two modulus? And what will be an advantage?

(c) Now do the same as in the previous exercise but take the following 3 rules into account:
- m and the offset c are relative prime.
- $a-1$ is divisible by all prime factors of m.
- $a-1$ is divisible by 4 if m is divisible by 4.

(d) Now construct the LSG described in Numerical Recipes, which has $a = 1664525$, $m = 2^{32}$ and $c = 1013904223$. Explain the result of this for the plot and the correlation coefficient.

(e) Now use `numpy.random` to generate a plot of random numbers and compare the previous results with this plot, also calculate the correlation.

4. **Cauchy Distribution**

Often one of the most typical distributions is the Cauchy distribution. This distribution is infamous for not having a defined expectation value and variance. Despite that, in astrophysics we are not afraid of the Cauchy distribution, and it is of significant importance in single star spectroscopy or spectroscopy in general. This has to do with the fact that the quantum-mechanical behaviour of a single spectroscopic line will result in a natural broadening of the line which corresponds to a Cauchy profile. In general the Cauchy distribution is given by:

$$f(x; x_0, \gamma) = \frac{1}{\pi\gamma\left(1 + \left(\frac{x-x_0}{\gamma}\right)^2\right)}. \tag{8}$$

In this exercise we will generate random numbers that will follow a Cauchy distribution. To generate the random numbers we will use (inverse) transform sampling. To do this the first thing we need to do is calculate the CDF or cumulative distribution function. This function is the probability that a random number is drawn below the current x, therefore mathematically the CDF is given by:

$$f_{\mathrm{CDF}}(x; x_0, \gamma) = \int_{-\infty}^{x} f(t; x_0, \gamma)\mathrm{d}t. \tag{9}$$

(a) Calculate the CDF for the Cauchy distribution.

(b) Now that we have the CDF we can use the properties of the CDF, given that is is between 0 and 1. This means that if we draw a random number between 0 and 1 we can generate a random number by inverting the CDF. Invert the CDF.

(c) The next step is using the derivation to calculate random numbers that follow a Cauchy distribution. Calculate random numbers and compare with the PDF.

5. **Von Neumann Rejection Sampling**
In this next part we are going to sample random numbers such that we can generate an actual line pattern of an emission or absorption line in a star or in a gas cloud in the ISM or IGM. To do this we first need to understand that there are basically three contributions to the profile of the line. The first is the quantum-mechanical behaviour of the line which produces a Cauchy profile, but besides this we have two other contributions to the line: thermal broadening of the line which produces a Gaussian profile for a Dirac delta line, and the velocity broadening of the line which similarly produces a Gaussian profile. This means we basically have a convolution of a Cauchy profile and a Gaussian profile, which is called a Voigt profile. The Voigt profile centered around zero is given by:

$$V(x;\sigma,\gamma) = \int_{-\infty}^{\infty} G(x';\sigma)L(x-x';\gamma)\mathrm{d}x'. \tag{10}$$

Here $G(x;\sigma)$ is a normalized centered Gaussian profile and $L(x,\gamma)$ is the centered normalized Cauchy profile.

(a) In general it is very hard to use inverse transform sampling on this function because the function already is a convolution. Because of this we will use the Von Neumann rejection technique (combining transform and rejection sampling). Use an uniform distribution to generate random numbers that follow a Voigt profile using rejection sampling.

(b) Do the same but now repeat your calculation by using the Cauchy distribution to calculate a distribution that is given by the Voigt profile. Use the calculation of the Cauchy distribution used in one of the previous exercises.

6. **Inversive Congruential Generator**
Besides the Linear Congruential generator there also are nonlinear congruential pseudo-random number generators like the inversive congruential generator. This random number generator uses another technique: it uses the modular multiplicative inverse compared to the previous value used. In general the rule of the inverse congruential generator is given by:

$$x_{i+1} = (ax_i^{-1} + c)\mathrm{mod}\ q \quad \text{if } x_i \neq 0, \tag{11}$$

$$x_{i+1} = c \quad \text{if } x_i = 0. \tag{12}$$

Make the same plots as in the case of the linear congruential generator and calculate the Pearson correlation coefficient for the following cases:

(a) Test for the case $a = 61$, $b = 2$ and $m = 256$.

(b) Secondly we will consider the case $a = 85$, $b = 2$ and $m = 256$.

(c) Now consider the case $a = 101$, $b = 2$ and $m = 256$.

(d) Lastly consider the case $a = 237$, $b = 2$ and $m = 256$.

(e) Compare your result with random numbers from `numpy.random` and compare with the LCG.